

Exploratory Data Analysis Report

Medical Appointment No-Show Prediction

Introduction

Missed medical appointments (no-shows) are a major source of lost hospital capacity, wasted staff time, and delayed care for other patients. This report presents the Exploratory Data Analysis (EDA) performed on a real-world hospital appointment dataset, with the goal of understanding patient demographics, medical conditions, and scheduling behaviour that relate to appointment attendance.

The dataset records details such as patient gender, age, neighbourhood, scholarship enrollment, chronic conditions (hypertension, diabetes, alcoholism), handicap level, whether an SMS reminder was sent, and whether the patient ultimately showed up (No-show).

2. Objectives of this EDA

- Understand the structure and distribution of each feature in the dataset.
- Identify and quantify the class imbalance in the target variable (No-show).
- Examine relationships between demographic/medical features and appointment attendance.
- Interpret ambiguous features (e.g., Handicap levels 0-4) with clearly justified assumptions.
- Identify confounding relationships (e.g., SMS_received and WaitingDay) before drawing conclusions.
- Provide a foundation for feature engineering and model selection in the next phase of the project.

3. Visualizations and Observations

1. Distribution of Appointment Attendance (Target Variable)

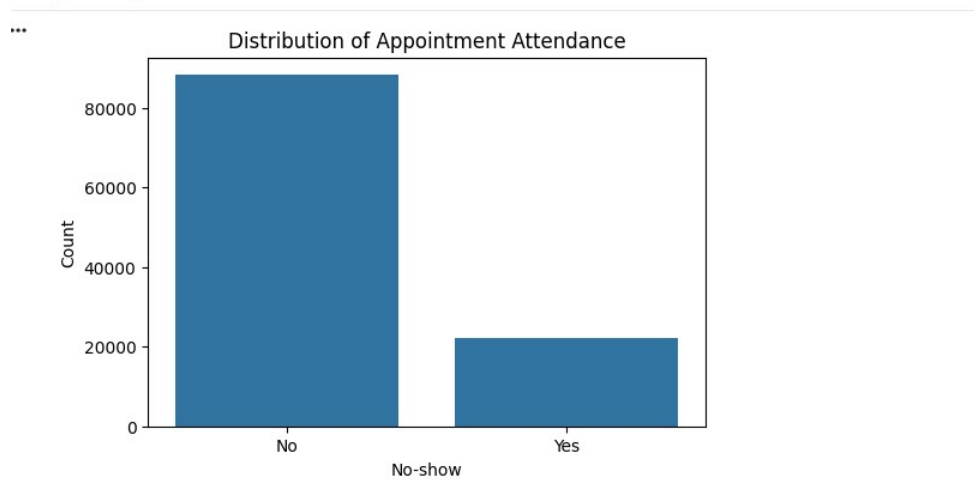


Figure 1: Distribution of Appointment Attendance (Target Variable)

Observation:

Approximately 88,000 patients attended their appointments while around 22,000 missed them, giving roughly an 80/20 split. This confirms a significant class imbalance in the target variable (No-show). Because of this imbalance, accuracy alone will not be a reliable evaluation metric during modelling, and techniques such as SMOTE (Synthetic Minority Oversampling) or class-weight balancing will be applied before training classifiers.

2. Age Distribution

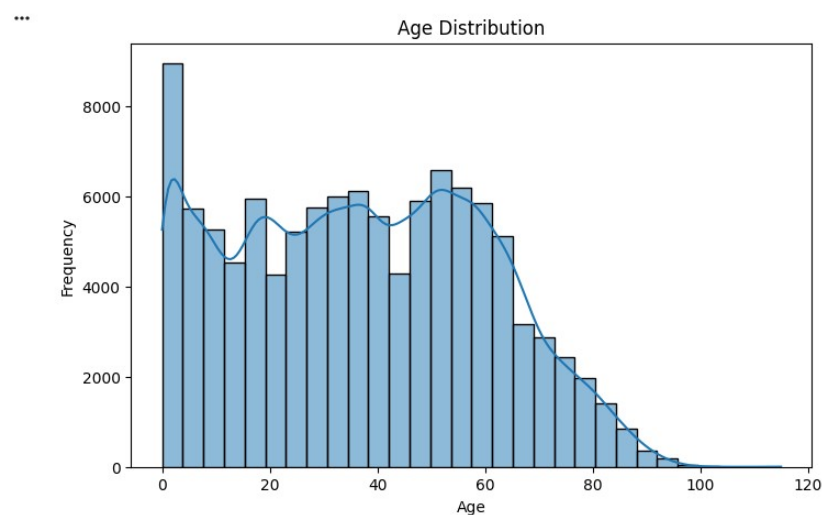


Figure 2: Age Distribution

Observation:

The age distribution is right-skewed. There is a large spike in the 0-5 age range, likely corresponding to infant vaccination and routine paediatric check-ups. The distribution is fairly spread across the 20-60 age range and gradually tapers off after age 80, reflecting the smaller proportion of elderly patients in the dataset.

3. Gender vs No-show

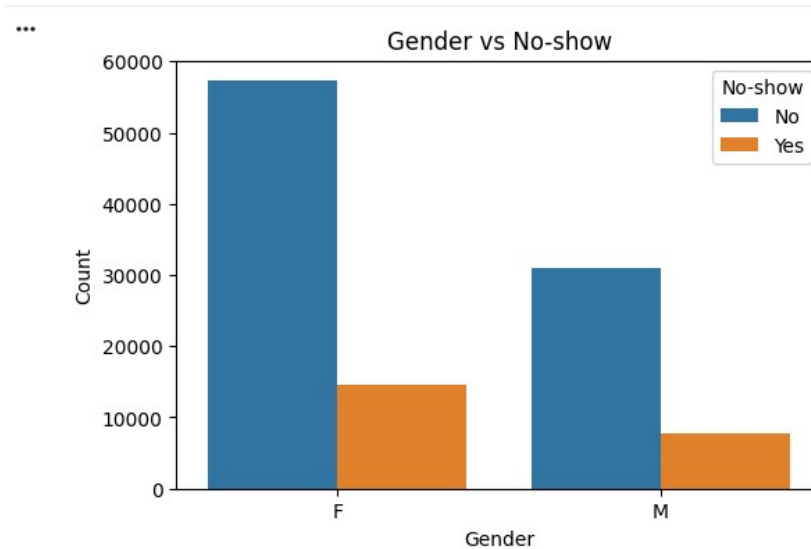


Figure 3: Gender vs No-show

Observation:

Female patients booked substantially more appointments than male patients (roughly double). However, the proportion of no-shows relative to total appointments is similar for both genders (around 20%), suggesting that gender alone is not a strong predictor of no-show behaviour.

4. Age vs No-show

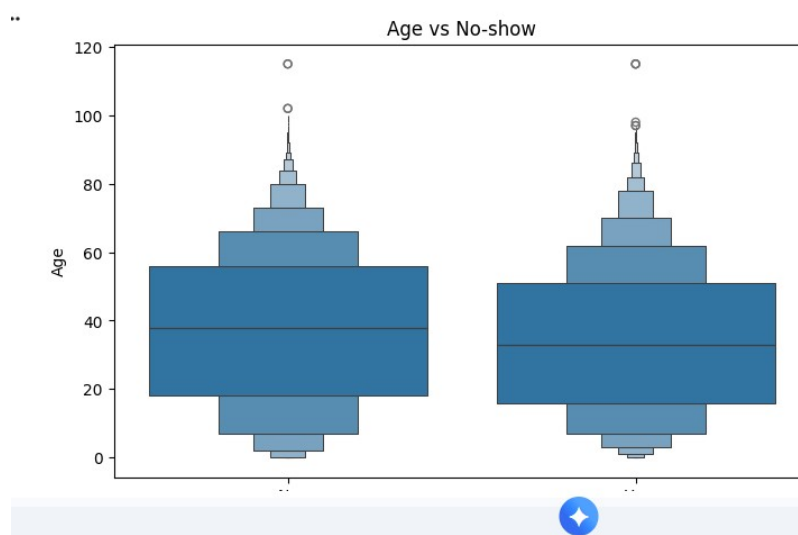


Figure 4: Age vs No-show

Observation:

Patients who attended their appointments have a slightly higher median age (~38) compared to patients who missed their appointments (~33). This suggests that younger patients are marginally more likely to miss scheduled appointments than older patients, though the difference is moderate rather than extreme.

5. Hypertension vs No-show

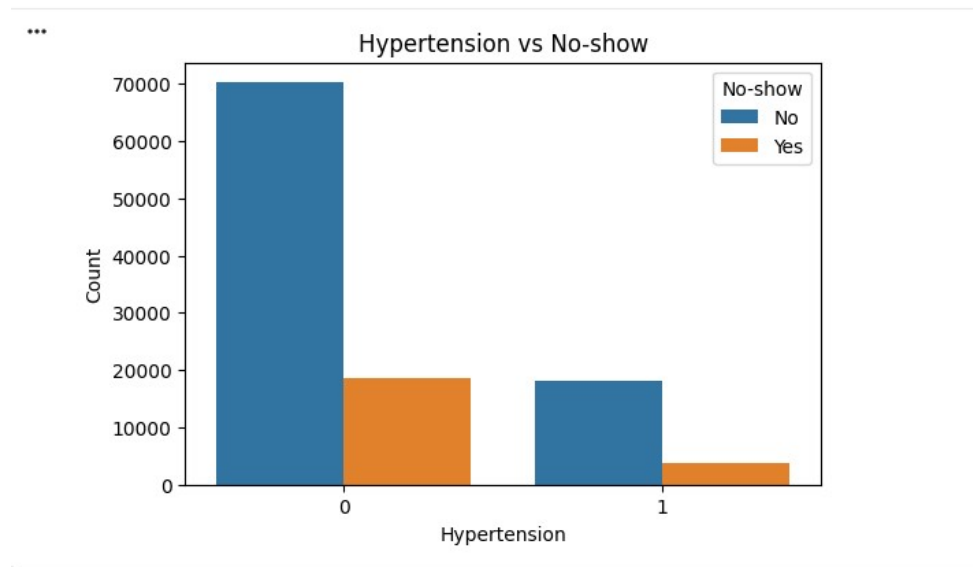


Figure 5: Hypertension vs No-show

Observation:

The proportion of no-shows is similar between patients with and without hypertension. This indicates that hypertension status by itself is not a strong differentiator of appointment attendance, though it may still contribute value in combination with other features.

6. Diabetes vs No-show

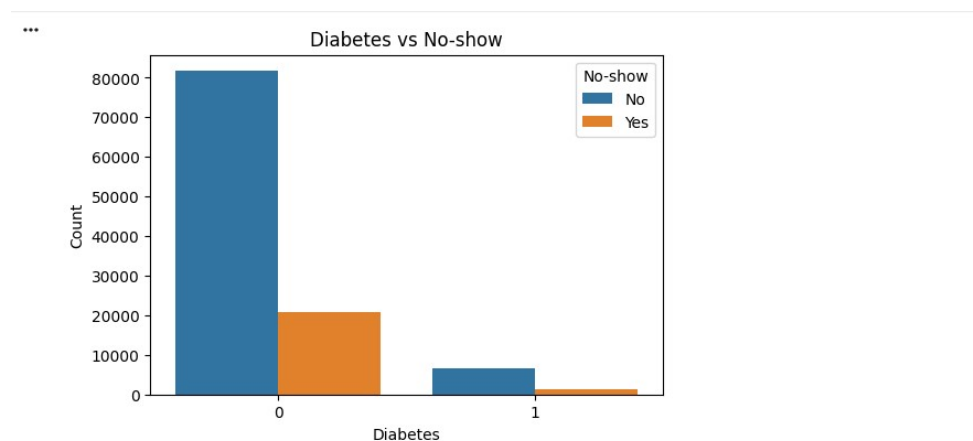


Figure 6: Diabetes vs No-show

Observation:

Similar to hypertension, the no-show proportion for diabetic and non-diabetic patients is nearly identical. This feature shows a weak isolated relationship with the target variable when examined independently.

7. Handicap vs No-show

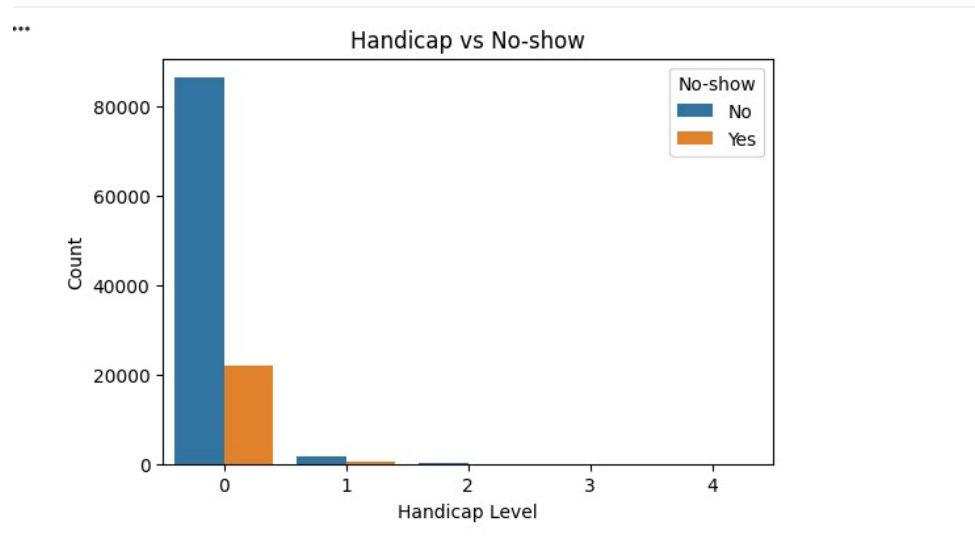


Figure 7: Handicap vs No-show

Observation:

The overwhelming majority of records have Handicap = 0, with very few records at levels 1-4. Because of this severe sparsity, the higher handicap levels do not carry enough data to draw statistically reliable conclusions. For modelling purposes, the Handicap feature was interpreted as an ordinal severity indicator (0 = none, 1-4 = increasing degrees of disability), and given the sparsity of levels 2-4, it may alternatively be simplified into a binary flag (Handicap = 0 vs Handicap > 0) to reduce noise -- this assumption is explicitly documented as a modelling decision.

8. Correlation Heatmap (Numeric Features)

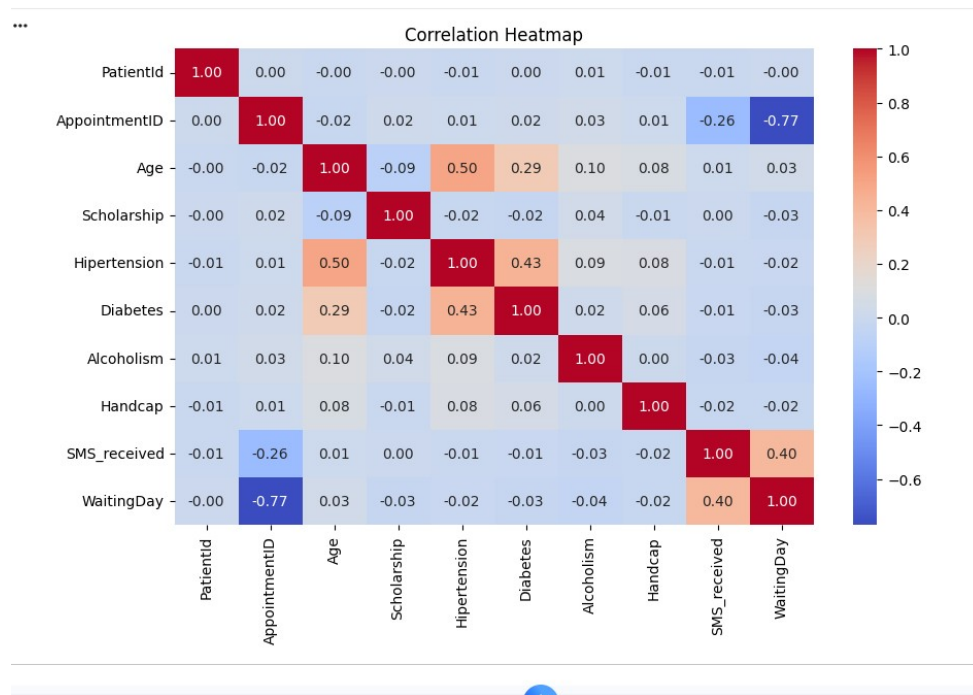


Figure 8: Correlation Heatmap (Numeric Features)

Observation:

Hypertension shows a moderate positive correlation with Age (0.50), which is expected since hypertension risk increases with age. Diabetes also correlates with Hypertension (0.43). SMS_received correlates with WaitingDay (0.40), meaning reminder SMS messages are more frequently sent for appointments booked further in advance. AppointmentID has a strong negative correlation with WaitingDay (-0.77), which is an artifact of ID sequencing rather than a meaningful clinical relationship. Importantly, no individual feature shows a strong direct correlation with No-show, suggesting the relationship is non-linear and better captured by tree-based models such as Random Forest or XGBoost rather than plain logistic regression.

9. Scholarship vs No-show

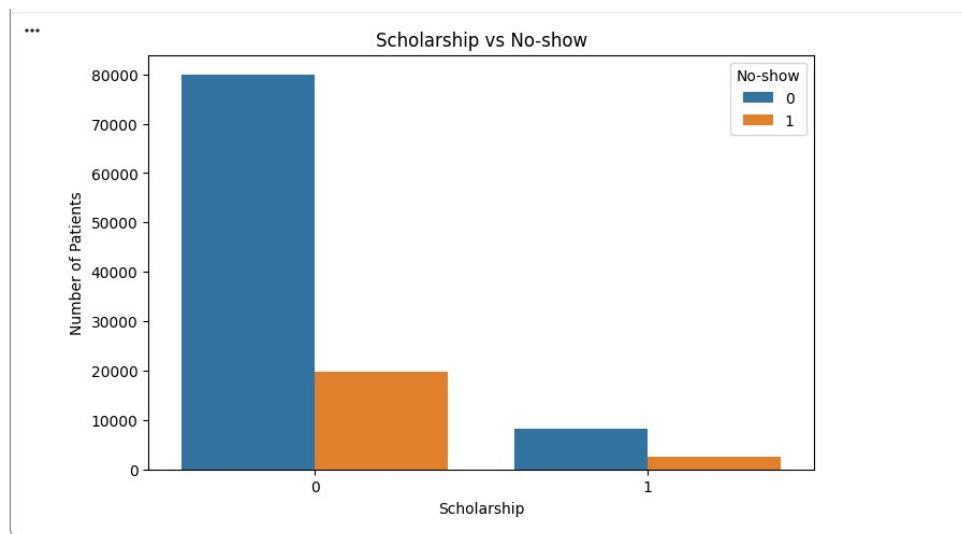
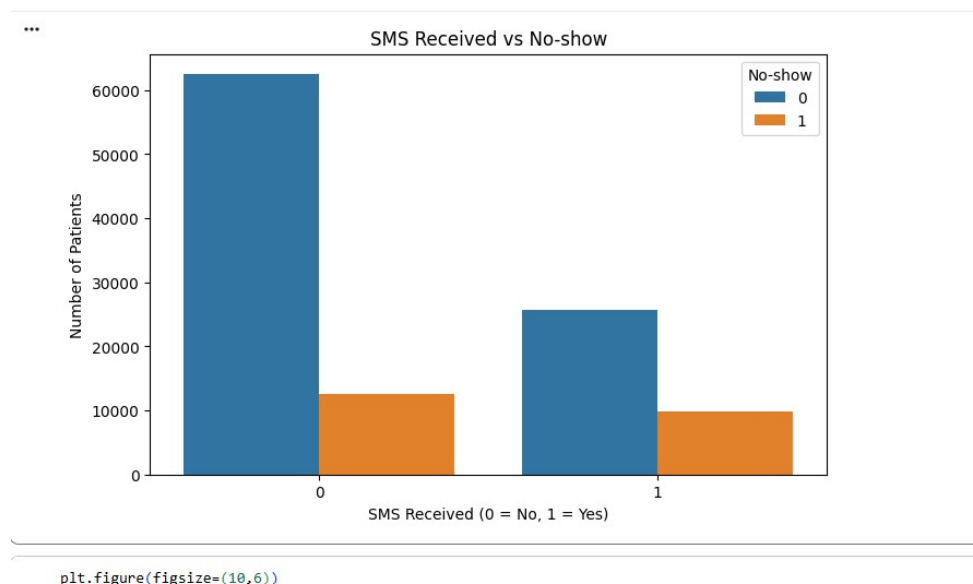


Figure 9: Scholarship vs No-show

Observation:

Patients without a scholarship (Bolsa Familia welfare enrollment) make up the majority of the dataset, and their no-show proportion is only marginally higher than that of scholarship holders. This suggests Scholarship is a weak standalone predictor.

10. SMS Received vs No-show



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plt.figure(figsize=(10,6))
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Figure 10: SMS Received vs No-show

Observation:

Counter-intuitively, patients who received an SMS reminder (1) show a higher no-show proportion (~27%) than patients who did not receive one (~20%). This does not mean SMS reminders cause no-shows. As seen in the correlation heatmap, SMS_received is positively correlated with WaitingDay (0.40) -- SMS reminders are mostly

sent for appointments booked far in advance, and longer waiting times are independently associated with higher no-show rates. This is a case of a confounding variable, and it is documented as such so the effect of SMS is not misinterpreted in isolation. Waiting time should be controlled for when assessing the true effect of SMS reminders.

11. Top 10 Neighbourhoods vs No-show

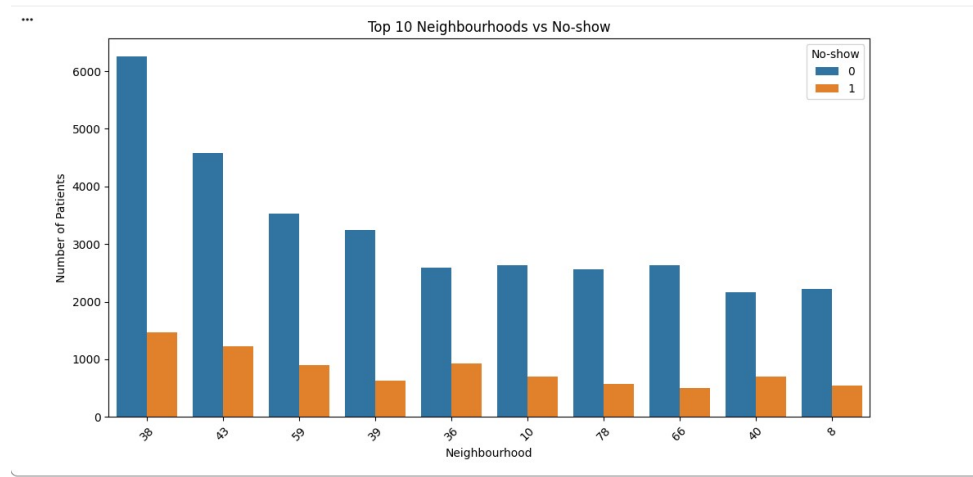


Figure 11: Top 10 Neighbourhoods vs No-show

Observation:

No-show proportions vary somewhat across neighbourhoods, with some areas showing a visibly higher share of missed appointments relative to their total volume. This indicates that location/neighbourhood may hold some predictive value and is worth retaining as a categorical feature (encoded appropriately) during feature engineering.